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Unveiling Solid-State Electrochemical Oxidation of $LiBH_4$ and $Li_2B_{12}H_{12}$ for High-Voltage All-Solid-State Batteries

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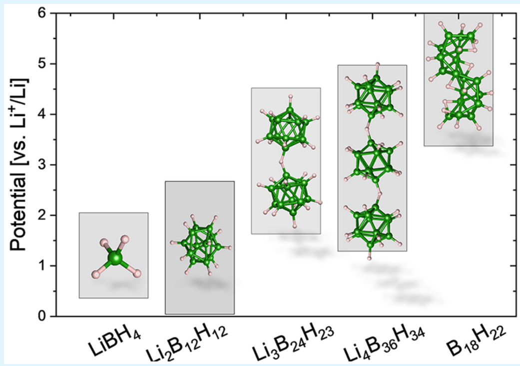
ABSTRACT: Hydridoborates are an emerging class of solid electrolytes that offer high ionic conductivity, low density, solution processability, compatibility with metallic anodes, and high oxidative stability. Notably, certain Li^+ or Na^+ solid electrolytes, consisting of two different cage-like *closo*-hydridoborate anion species, are compatible with 4 V-class cathodes by forming a sufficiently ion-conductive, passivating interphase. However, the nature of their electrochemical decomposition products and their dependence on electrochemical potentials remain unclear. In this combined theoretical and experimental study, we demonstrate the solid-state electrochemical oxidation of $LiBH_4$ to $Li_2B_{12}H_{12}$ above 2.0 V vs Li^+/Li and provide evidence for the successive oxidation of *closo*-$[B_{12}H_{12}]^{2-}$ anions to larger H-interconnected *closo*-clusters. This supports the observed trend that larger clusters formed via oxidation are stabilized at higher electrochemical potentials. Notably, the oxidation process from $LiBH_4$ to $Li_2B_{12}H_{12}$ proceeds through the formation of a highly conductive $[BH_4]^-$−$[B_{12}H_{12}]^{2-}$ mixed phase, indicating the potential for *in situ* formation of mixed-anion hydridoborates directly within all-solid-state cells. These insights into solid-state electrochemical decomposition at the solid−solid interfaces are transferable to other hydridoborate systems, regardless of cation species or anion structures, contributing to developing cathode design strategies for high-voltage all-solid-state batteries.



KEYWORDS: *all-solid-state batteries, solid electrolytes, lithium-ion conductors, hydridoborates, borohydrides, first-principles calculations, electrochemical stability*

1. INTRODUCTION

Solid electrolytes with liquid-like ionic conductivities, stability against alkali metal anodes, and sufficiently high oxidative stability to allow the use of high-voltage cathodes are prerequisites for all-solid-state batteries with high power and energy density.[1,2] To compete with the latest lithium-ion battery technology, electrolytes must also be economical, environmentally friendly, easy to process, lightweight, and have low gravimetric density. Alkali metal hydridoborates, formerly known as hydroborates, are a promising class of solid electrolytes that meet most of the above requirements. In particular, the lithium and sodium salts of *closo*- (= cage-like) hydridoborate $[B_nH_n]^{2-}$ or monocarba-hydridoborate $[CB_{n-1}H_n]^-$ (n = 10,12) anions combine liquid-like ionic conductivities (>10^{-3} S cm^{-1}) with a cation transference number of almost unity, compatibility with lithium and sodium metal anodes, high oxidative stability (>3 V vs Li^+/Li and Na^+/Na), low gravimetric densities (≤1.2 g cm^{-3}), high thermal and chemical stability, soft mechanical properties allowing cold pressing, solution processability, and low toxicity.[3−6]

Within the family of hydridoborates, $LiBH_4$ is the most studied hydridoborate and serves as a prototypical example since its superionic conductivity was first discovered in 2007.[7] Its lithium-ion conductivity rises abruptly to >10^{-3} S cm^{-1} above 110 °C due to a first-order transition from an orthorhombic low-temperature symmetry to a hexagonal high-temperature symmetry.[7] Its high-temperature phase can be stabilized at room temperature by the incorporation of anions such as halides (i.e., in a solid solution with LiX (X = Cl, Br, I))[8,9] or amide, reaching ionic conductivities of up to 6.4 · 10^{-3} S cm^{-1} near room temperature (40 °C) in $[BH_4]^-$-rich $Li(BH_4)_{1-x}(NH_2)_x$ compounds.[10] All-solid-state cells with $LiBH_4$-based solid electrolytes have been realized by coupling with electrodes such as lithium metal, $Li_4Ti_5O_{12}$, TiS_2, and sulfur.[10−12] These cells demonstrate ionic transport through $LiBH_4$ as a solid electrolyte; however, the cell voltage without a protection layer is limited up to 2.5 V, mainly due to the narrow electrochemical stability window of $LiBH_4$ between 0.7 and 2.1 V vs Li^+/Li.[13,14] Its thermodynamic instability with

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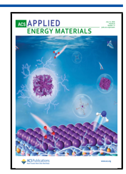

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lithium metal anodes (= 0 V vs Li^+/Li) induces interphase formation at the electrolyte/anode interface,[15] while the direct evidence of electrochemical $LiBH_4$ oxidation at the electrolyte/cathode interface beyond its oxidative stability limit is missing.

$Li_2B_{12}H_{12}$ is a lithium salt of the *closo*-hydridoborate anion $[B_{12}H_{12}]^{2-}$ and is known as one of the main thermal decomposition products of $LiBH_4$, whose decomposition pathway depends on both the applied temperature and the hydrogen pressure.[16] As described above for $LiBH_4$, $Li_2B_{12}H_{12}$ also undergoes a transition to its high-temperature phase at 355 °C, accompanied by an expansion of the unit cell and a facilitated anionic reorientation motion.[17,18] At room temperature, $Li_2B_{12}H_{12}$ exhibits the ionic conductivity of only $5 \cdot 10^{-8}$ S cm^{-1},[19] comparable to the room-temperature ionic conductivity of $LiBH_4$.[7] As in the case of $LiBH_4$, anion mixing is the method of choice to stabilize a highly conductive phase at room temperature. A single-phase equimolar mixture of two species of *closo*-hydridoborates, $Li_2B_{10}H_{10}$ and $Li_2B_{12}H_{12}$, enhances its ionic conductivity by several orders of magnitude compared to its respective anion components and exhibits $4 \cdot 10^{-4}$ S cm^{-1} at 25 °C and $4 \cdot 10^{-3}$ S cm^{-1} at 60 °C.[19] Even higher room-temperature ionic conductivities between $6.9 \cdot 10^{-5}$ S cm^{-1} and $6.7 \cdot 10^{-3}$ S cm^{-1} have been reported for mixtures of *closo*-monocarba-hydridoborates $(1-x)LiCB_9H_{10}-xLiCB_{11}H_{12}$ (x = 0.1−0.9).[20,21] On the other hand, sodium salts of *closo*-hydridoborates, *closo*-monocarba-hydridoborates, and their mixtures generally have an even higher ionic conductivity than lithium salts,[22−25] which peaks in $Na_2(CB_{11}H_{12})(CB_9H_{10})$ with $7 \cdot 10^{-2}$ S cm^{-1} at 27 °C.[26] Clear comparative compilations of ionic conductivities of selected hydridoborates can be found in previous studies.[5,19,20]

Moreover, the higher stability of compounds containing $[B_nH_n]^{2-}$ and/or $[CB_{n-1}H_n]^-$ (n = 10, 12) anions, compared to $LiBH_4$, has been confirmed by the evaluation of their electrochemical stability windows, typically exceeding 3 V vs Li^+/Li.[14,19,22,24,27−29] Such high intrinsic stability of *closo*-hydridoborate-based solid electrolytes has enabled the successful integration into 3 V and even 4 V all-solid-state battery cells with stable cycling against alkali metal anodes, e.g. Li|$Li_3(CB_{11}H_{12})_2(CB_9H_{10})$| $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$, Na|$Na_4(B_{10}H_{10})_2(B_{12}H_{12})$|$NaCrO_2$, and Na|$Na_4(CB_{11}H_{12})_2(B_{12}H_{12})$|$Na_3(VOPO_4)_2F$.[28,30,31] Mixing anions has only a minor effect on the stability of the constituent anions.[27,28] However, mixing a more stable anion with a less stable anion may lead to the kinetic stabilization or the formation of a stable interphase, which allows the cell operation beyond the stability limit of the less stable anion.[28] Notably, in Na|$Na_4(CB_{11}H_{12})_2(B_{12}H_{12})$|$Na_3(VOPO_4)_2F$, the partial decomposition of the $[B_{12}H_{12}]^{2-}$ anions (oxidative stability limit of $Na_2B_{12}H_{12}$: ~3.5 V vs Na^+/Na), forms a passivation layer in the cathode composite using a 4 V-class cathode, which allowed stable cycling up to 4.15 V without an additional cathode protection layer.[28,32] The chemical nature of the passivating interphase is under question. Dimer formation was proposed by analogy to electrochemical oxidation of $[B_{10}H_{10}]^{2-}$ or $[B_{12}H_{12}]^{2-}$ anions in solution.[33−37] Recent studies on sodium *closo*-hydridoborates show extended oxidative stability of $[B_{10}H_{10}]^{2-}$ and $[B_{12}H_{12}]^{2-}$ dimers, and a $[B_{12}H_{12}]^{2-}$ trimer, which can also be employed as a building block for anion mixing.[38−40] However, the dimers/trimers used in these works were all prepared by chemical or thermal decomposition of $[B_{10}H_{10}]^{2-}$ or $[B_{12}H_{12}]^{2-}$ anions in aqueous solutions;[33,41,42] therefore, the electrochemical decomposition pathway of *closo*-hydridoborates in all-solid-state cells remains unknown.

Here we combine first-principles calculations and experiments to elucidate the solid-state electrochemical oxidation of $LiBH_4$ and $Li_2B_{12}H_{12}$, unveiling the solid-state electrochemical oxidation pathway of $[BH_4]^-$ to $[B_{12}H_{12}]^{2-}$ and further to longer H-interconnected *closo*-clusters. We also propose the formation of a highly conductive $[BH_4]^-{-}[B_{12}H_{12}]^{2-}$ mixed phase from the partial decomposition of $LiBH_4$, exemplifying a route to the *in situ* preparation of mixed-anion hydridoborates. The generic electrochemical oxidation behavior of hydridoborate anions, supported by theory and experiment, will facilitate the design of cathodes for high-voltage all-solid-state batteries.

2. THEORETICAL AND EXPERIMENTAL METHODS

2.1. First-Principles Calculations. The electrochemical stability window calculations were performed using the Vienna ab initio simulation package (VASP) within periodic density functional theory (DFT).[43] The atoms were represented with the projector augmented wave (PAW) potentials[44,45] with the valence configuration of $1s^1$ for H, $1s^22s^1$ for Li, and $2s^2\,2p^1$ for B. The gradient-corrected exchange−correlation functional, according to Perdew, Burke, Ernzerhof (PBE),[46] and the nonlocal correction accounting for weak dispersive interaction were applied.[47] The plane wave basis set energy cutoff was 700 eV. For calculations of the molecules, a large periodic box with at least 9 Å of vacuum-separated periodic images was implemented. This computational methodology was applied to investigate the electrochemical stability window or interfacial stability of solid electrolytes, which is typically calculated using the grand canonical phase diagram method.[13,48] This method is based on the principle that a solid electrolyte decomposes into a series of stable phases or compounds with overall stoichiometry corresponding to the decomposition reaction, e.g. delithiation for oxidation at the cathode or lithiation for reduction at the anode. Thus, this method provides upper and lower stability limits of electrolytes and serves as a guideline for assessing the electrochemical stability of new solid electrolytes. However, the predicted stability window is often narrower than those observed in experiments.[49,50]

This discrepancy arises because thermodynamic stability does not account for kinetic barriers that may hinder specific reactions. Additionally, this method relies on known, thermodynamically stable compounds that form crystalline structures, while electrolyte decomposition typically occurs at the electrolyte interface with active materials and/or electronic conductive additives in the electrode. At these interfaces, the formation of an interfacial layer is common, and such layers are often poorly understood. Li-poor or Li-rich phases at these interfaces are often off-stoichiometric and do not need to consist of thermodynamically stable phases when isolated. Accounting for these facts brings an excellent match between predictions and observations in NASICON-type solid electrolytes.[50]

Neutral hydridoboranes and hydridoborate anions present additional challenges because B−H compounds form a very large variety of structures, including *closo*-, *nido*-, *arachno*-, and *hypo*-clusters,[51] all of which possess multicenter bonding and are usually aromatic. These complications were indirectly noticed by the simple omission of the Li-poor composition formed at the cathode, which provides a better agreement between calculations and measurements[13,14]

In this work, we propose a method that provides insight into the stability of the oxidation products of $LiBH_4$ and $Li_2B_{12}H_{12}$. Our approach considers polymeric chains of *closo*-anions and addresses noncrystalline structures approximated with atomic and molecular clusters. The details of this method are outlined in the Supporting Information, which includes an accuracy assessment (Figures S1 and S4), the range of possible decomposition products (Figure S2), stable compounds at relevant electrochemical potentials (Figure S3), and the bonding properties of the anions (Figure S5). The electrochemical stability of $[B_{12}H_{12}]^{2-}$ anions is predominantly governed by intramolecular bonding, while the electrostatic and dispersive interaction between anions[52] can already be approximated for clusters of just two molecules. This allows stability calculations of nonperiodic structures (Figure S4).

2.2. Material Preparation. Commercial $LiBH_4$ (95%, Sigma-Aldrich) was ground for 15 min with mortar and pestle and used without further purification. $Li_2B_{12}H_{12}\cdot 4H_2O$ and $Li_3B_{24}H_{23}\cdot xH_2O$ were purchased from KatChem. As-received $Li_2B_{12}H_{12}\cdot 4H_2O$ was vacuum-dried ($p < 10^{-3}$ mbar) at 210 °C for 14 h using Schlenk technique; as-received $Li_3B_{24}H_{23}\cdot xH_2O$ was vacuum-dried ($p < 10^{-3}$ mbar) at 150 °C for 12 h after the determination of suitable drying conditions by combined differential scanning calorimetry (DSC), thermogravimetry (TG), and mass spectrometry (MS) (Netsch STA449 F3 Jupiter with QMS 403C Aëolos) (see Figure S6). For electrochemical stability measurements, dried $Li_2B_{12}H_{12}$ was ball-milled for 3 × 15 min in a 1:30 sample-to-ball weight ratio, using a shaker mill (Spex 8000M) to enhance its ionic conductivity.[53,54]

Carbon black with a bulk density ρ_{bulk} of 0.16 g cm^{-3} (Super C65, Imerys Graphite & Carbon) was vacuum-dried ($p < 10^{-3}$ mbar) at 150 °C for 12 h and used as an electronically conductive additive to accelerate the kinetics of electrochemical decomposition reactions.[27] The BET specific surface area S_{BET} of the dried Super C65 was measured to be 60 m^2 g^{-1} by five-point nitrogen adsorption, using a surface area and porosity analyzer (Micromeritics ASAP 2020). All chemicals were stored and handled in an argon-filled glovebox (MBraun) with O_2 and H_2O levels below 0.1 ppm.

2.3. Material Characterization. Combined DSC-TG-MS measurements were performed under helium flow at a heating/cooling rate of 5 K min^{-1} (Netsch STA 449 F3 Jupiter with QMS 403C Aëolos). Samples (typically 5−10 mg) were sealed in aluminum pans in the glovebox under inert atmosphere. The pans were punctured and immediately transferred to the sample stage prior to the DCS-TG-MS measurements to minimize air contact.

Infrared spectroscopy (IR) spectra were acquired between 600 cm^{-1} and 4000 cm^{-1} with a resolution of 1 cm^{-1} (Bruker Alpha FT-IR spectrometer in attenuated total reflection configuration).

Nuclear magnetic resonance (NMR) measurements were performed on a 5 mm cryoprobe (Bruker Avance III 400 NMR spectrometer with CryoProbe Prodigy). Samples (typically 2−7 mg) collected from the working electrode of the electrochemical decomposition measurements were dissolved in 1 mL anhydrous acetonitrile-d_3 or DMSO-d_6 (Sigma-Aldrich) and filled in borosilicate glass tubes in a glovebox under inert atmosphere. Single pulse ^{11}B NMR spectra were recorded at 25 °C at 128.4 MHz with ^{1}H decoupling during acquisition, applying 30° pulses of 5 μs and recycle times of 5.3 s, which enables quantitative recording of the spectra (typically 512 scans were recorded). The ^{11}B chemical shifts (δ) in ppm were calibrated to an external sample of 35% BF_3 diethyl etherate in $CDCl_3$ solution at 0.0 ppm. The boron background resonance of the probe was subtracted from the spectra with data recorded under the same measurement conditions from sample-free solvents.

X-ray photoelectron spectroscopy (XPS) measurements were performed at room temperature with a Ulvac-Phi Quantum 2000, using a monochromatic Al Kα X-ray source (1486.6 eV) with a pass energy of 30 eV. All measurements were conducted on three different areas with 150 μm diameter for each sample. Sample charging was prevented by charge compensation provided by a low-energy electron and an argon ion gun. XPS data were processed with the CasaXPS software. Spectra were calibrated by setting the hydrocarbon component of the C 1s photoemission peak to 285.0 eV binding energy.

Synchrotron XRD patterns were collected in the transmission mode with a wavelength of 0.7095 Å at room temperature, using a Mythen detector at the MS-X04SA beamline of the Swiss light source (SLS) at the Paul Scherrer Institute in Villigen, Switzerland.[55] Powder samples were ground with a mortar and pestle, and filled and sealed in borosilicate glass capillaries in a glovebox under inert atmosphere. Phase identifications of XRD patterns were performed by the HighScore software.

2.4. Electrochemical Characterization. Following our previously established method to determine the electrochemical oxidative stability of hydroborates, $LiBH_4$ or ball-milled $Li_2B_{12}H_{12}$ was mixed with Super C65 in a weight ratio of 75:25 for 15 min with mortar and pestle.[27] For the electrochemical decomposition measurements, a powder stack of (i) a 5 mg layer of the $LiBH_4$/Super C65 composite and (ii) an 80 mg electrolyte layer of $LiBH_4$ was uniaxially pressed under 450 MPa to form a pellet with 12 mm diameter and 1 mm thickness. A platinum disk (99.997%, 0.25 mm in thickness, Alfa Aesar) and an aluminum foil (>99.3%, 15 μm in thickness, MTI) with 12 mm diameter (1.13 cm^2) were attached to the carbon composite side of the pellet. A surface-scratched lithium metal foil (99.9%, 0.5 mm in thickness, China Energy Lithium) with 10 mm diameter (0.785 cm^2) was attached to the other side. Following the same procedure as described above, the electrochemical oxidation of $Li_2B_{12}H_{12}$ was carried out to 5 mg of the $Li_2B_{12}H_{12}$/Super C65 composite and 80 mg of ball-milled $Li_2B_{12}H_{12}$. A uniaxial pressure of 150 MPa was applied to avoid pellet cracking. Electrodes were attached to the pellet as described above.

A multichannel potentiostat (BioLogic VMP3) was used for electrochemical characterization. For electrochemical stability measurements, cyclic voltammetry was performed at a scan rate of 10 mV s^{-1} at 120 °C between 1.5 and 4.0 V vs Li^+/Li for $LiBH_4$ and between 2.7 and 6.0 V vs Li^+/Li for $Li_2B_{12}H_{12}$. Galvanostatic measurements were performed at 120 °C at a constant current density of 48 μA cm^{-2} from open-circuit voltage (OCV) to 3.0 V vs Li^+/Li for $LiBH_4$, and at 15 μA cm^{-2} from OCV to 5.0 V vs Li^+/Li for $Li_2B_{12}H_{12}$. When the cell voltage reached each upper cutoff voltage, the voltage was held constant until the current density dropped to <15% of each applied constant current density to complete electrochemical reactions. Electrochemical impedance spectroscopy measurements were conducted in a frequency range from 100 kHz to 1 Hz with a 10 mV amplitude.

3. RESULTS AND DISCUSSIONS

3.1. Electrochemical Stability Window Calculations. Calculated electrochemical stability windows are presented in Figure 1a. While the $[BH_4]^-$ ion is the smallest hydridoborate anion, the $[B_{12}H_{12}]^{2-}$ ion is the most stable among the *closo*-hydridoborate anions $[B_nH_n]^{2-}$ (n = 12 for $[B_{12}H_{12}]^{2-}$).[56] $LiBH_4$ is stable between 0.64 and 2.05 V vs Li^+/Li, and its stability window is in agreement with previous studies.[13,14] Upon oxidation, $LiBH_4$ decomposes to $Li_2B_{12}H_{12}$ as is the case of thermal decomposition, which polymerizes into $[B_{24}H_{23}]^{3-}$ or $[B_{36}H_{34}]^{4-}$ polyanions at 2.4−2.7 V vs Li^+/Li (see Supporting Information for details). Since $[B_{36}H_{34}]^{4-}$ is more stable than $[B_{24}H_{23}]^{3-}$, only the $[B_{36}H_{34}]^{4-}$ will be present in the system on a purely thermodynamic basis. Therefore, the stability limits for $[B_{24}H_{23}]^{3-}$ are presented in a lighter shade (see Figure 1a). They are calculated with the assumption that polymerization does not proceed beyond $[B_{24}H_{23}]^{3-}$, which is treated as a metastable anion. The stability range of both polyanions extends from 2.4−2.7 V to 4.5−5.0 V vs Li^+/Li until the structure of *closo*-B_{12} cages is destroyed upon final decomposition (see Figure 1b). We chose to calculate $B_{18}H_{22}$ as a potential solid decomposition product because (i) it is neutral with no Li^+ cations and hence cannot be further oxidized by electrochemical delithiation, (ii) it is formed by two *closo*-B_{12} cage fragments sharing two B atoms without a major rearrangement of *closo*-B_{12} cages, and (iii) it was reported as an ultimate decomposition product in previous computational studies.[14]

The $[B_{24}H_{23}]^{3-}$ ion is a dimer of two $[B_{12}H_{12}]^{2-}$ ions, where two B_{12} cages are interconnected with one B−H−B bridging bond (see Figure 1b) that is formed by a loss of one H atom upon the dimerization reaction.[35−37,41,42,57] Likewise, the $[B_{36}H_{34}]^{4-}$ ion is a trimer of $[B_{12}H_{12}]^{2-}$ ions, where three B_{12} cages are interconnected linearly with two B−H−B

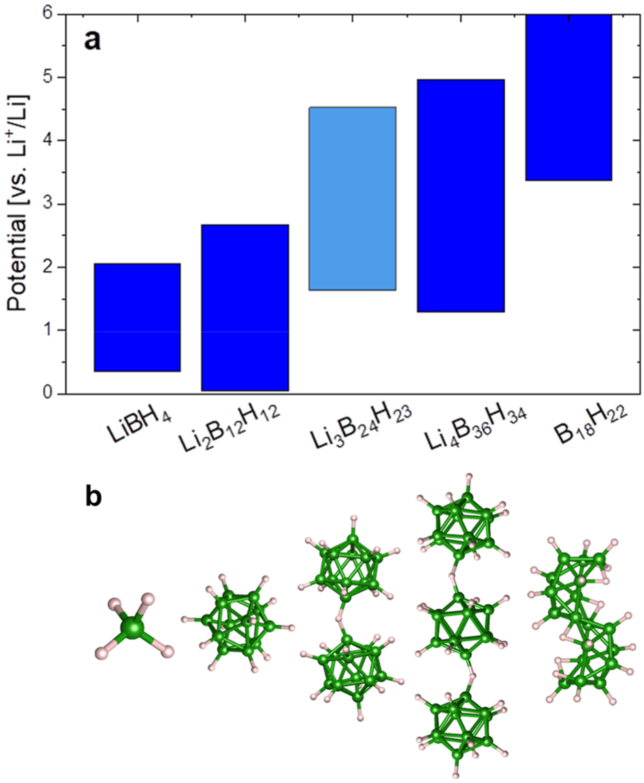

Figure 1. (a) The calculated electrochemical stability windows of $LiBH_4$ (0.37−2.05 V vs Li^+/Li), $Li_2B_{12}H_{12}$ (0.04−2.67 V vs Li^+/Li), $Li_3B_{24}H_{23}$ (1.69−4.52 V vs Li^+/Li), $Li_4B_{36}H_{34}$ (1.35−4.97 V vs Li^+/Li), and $B_{18}H_{22}$ (above 3.37 V vs Li^+/Li) by grand-potential phase diagrams. All potentials are given in relation to Li^+/Li. The light blue shade indicates that $Li_3B_{24}H_{22}$is metastable. (b) Schematic representation of $[BH_4]^-$, $[B_{12}H_{12}]^{2-}$ anions and of the oxidation products: $[B_{24}H_{23}]^{3-}$, $[B_{36}H_{34}]^{4-}$ polyanions, as well as delithiated $B_{18}H_{22}$. For details of the decomposition reactions, refer to Supporting Information.

bridging bonds.[42] These transformations do not require significant changes of the bonding orbitals in the *closo*-hydridoborate molecule; only orbital degeneracy is lifted, and additional bonds between molecules are created (see Figure S5). We also note that previous calculations of the electrochemical stability windows for hydridoborates[13,14] did not account for weak dispersive interactions that are crucial for aromatic molecules. Figure S4 presents that the underestimation of the electrochemical stability for $Li_2B_{12}H_{12}$ is partly related to these interactions.

To sum up, the grand canonical stability calculations are in line with the following electrochemical oxidation pathway:

$LiBH_4 \rightarrow Li_2B_{12}H_{12}$ above 2.0 V vs Li^+/Li

$\rightarrow$ dimer/trimer above 2.4−2.7 V vs Li^+/Li

$\rightarrow$ decomposition to $B_{18}H_{22}$ at 4.5−5.0 V vs Li^+/Li

The vibrational analysis performed for all polymorphs do not show any instabilities.

3.2. Electrochemical Oxidation of $LiBH_4$. To experimentally confirm the oxidative decomposition pathways suggested by electrochemical stability window calculations, we prepared a $LiBH_4$-carbon composite as a working electrode and applied constant current in a Li/$LiBH_4$/$LiBH_4$-carbon/Pt cell at 120 °C.[27] Figure 2a shows the galvanostatic curve of the cell and the subsequent constant voltage step at 3.0 V vs Li^+/Li. As previously reported, the voltage plateau starting at around 2.0 V vs Li^+/Li corresponds to the onset of anodic (= oxidative) currents in the cyclic voltammogram measured in the same cell setup at 120 °C (see Figure S7a), corroborating the electrochemical oxidation of $LiBH_4$.[27] The corresponding impedance spectra before the galvanostatic charge and after the constant voltage step are shown in Figure 2b. The real part of the impedance intersecting with the *x*-axis at high frequencies dropped by >30%, indicating that the electrochemical oxidation of $LiBH_4$ decreased the bulk impedance. This method is designed to determine the intrinsic electrochemical stability of the solid electrolyte in a composite with a high surface-area carbon conductive additive. In a practical cell configuration involving an active material, additional chemical reactions may occur at the interface, and kinetic limitations may differ due to additional interfaces or catalytic effects. Therefore, prior to integration with an active material, its compatibility must be carefully assessed based on the intrinsic electrochemical stability window, and, if necessary, protective measures should be implemented as discussed above.

The electrochemically oxidized composite in the working electrode of the cell was collected for further characterization by IR, ^{11}B NMR, and XRD, as shown in Figures 2c,d, and S8. By comparison to the reference materials, the $LiBH_4$ composite after electrochemical oxidation turns out to be a mixture of the newly formed $Li_2B_{12}H_{12}$ with residual $LiBH_4$. This provides clear experimental evidence that $Li_2B_{12}H_{12}$ is among the electrochemical oxidation products of $LiBH_4$ as in the case of thermal decomposition.[16,58−60] Also, this result is in line with former first-principles studies on the stability of $LiBH_4$,[13] supporting the validity of the calculated electrochemical stability window of $LiBH_4$ in this work, which considers $Li_2B_{12}H_{12}$ as a decomposition product above the upper stability limit. Moreover, the reduced bulk impedance after the electrochemical oxidation of $LiBH_4$ can be attributed to the increase in ionic conductivity upon the formation of $Li_2B_{12}H_{12}$. Although $Li_2B_{12}H_{12}$ itself exhibits lower ionic conductivity than that of $LiBH_4$,[7,19,54] the observed mixture of $LiBH_4$ and $Li_2B_{12}H_{12}$ may exhibit enhanced ionic conductivity, as observed previously.[61] Therefore, when $LiBH_4$ is used as a solid electrolyte, its oxidative decomposition above 2.0 V vs Li^+/Li would not lead to the cell resistance increase and degradation due to the stable interphase formation of $Li_2B_{12}H_{12}$. This enables stable cell cycling beyond the electrochemical stability window of $LiBH_4$ as long as the potential range is within the electrochemical stability window of the formed $Li_2B_{12}H_{12}$, corroborating previous studies.[11] It is worth noting that mixed-anion hydridoborates can retain their ionic conductivity even after extended cycling. Refs 28 and 30 report stable performance over hundreds to thousands of cycles, as long as the operating conditions remain within the stability limits of solid electrolytes.

In summary, $LiBH_4$ undergoes conversion to $Li_2B_{12}H_{12}$ through electrochemical oxidation in the solid state, a process corroborated by calculations of the electrochemical stability window. Unlike the suggested in thermal decomposition,[16,58−60] LiH is less likely to form since hydride anions (H^-) in LiH would also be oxidized to H_2 above 1.0 V vs Li^+/Li (see Figure S3).[13] Therefore, if we only assume H_2 gas evolution as another product in solid-state electrochemical reactions, the possible oxidation reaction of $[BH_4]^-$ anions can be written as

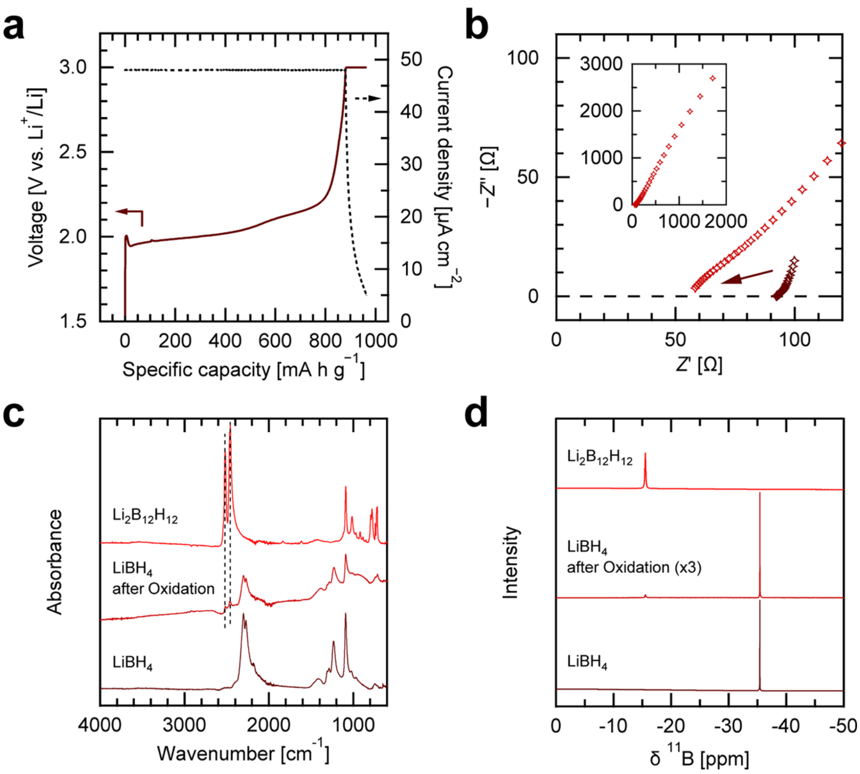

Figure 2. (a) Galvanostatic charge curve of electrochemical $LiBH_4$ oxidation at a current density of 48 μA cm^{-2} in a $Li/LiBH_4/LiBH_4$-carbon/Pt cell ($LiBH_4$:carbon = 75:25 in weight) at 120 °C. When the cell voltage reached 3.0 V vs Li^+/Li, the voltage was held constant until the current density dropped to <15% of the applied constant current density. The specific capacity was calculated based on the weight of $LiBH_4$ in the composite (= 3.75 mg). (b) Impedance spectra recorded before the galvanostatic charge (brown) and after the constant voltage step (dark red). The inset in (b) shows the impedance spectra over the full frequency range. (c) IR and (d) ^{11}B NMR spectra of $LiBH_4$ (brown), $LiBH_4$ after electrochemical oxidation (crimson), and $Li_2B_{12}H_{12}$ (red).

$$12[BH_4]^- \rightarrow [B_{12}H_{12}]^{2-} + 18H_2 + 10e^- \tag{1}$$

It is important to note that during electrochemical oxidation, an equal number of Li^+ ions and electrons are extracted from the solid electrolyte in the working electrode and transferred toward the counter electrode. This oxidation occurs as a side reaction during the delithiation reaction of the cathode active material upon charging when applied electrochemical potentials are beyond the oxidative stability limit of the electrolyte.

Eq 1 implies that safe operation of a $LiBH_4$-based battery system requires the prevention of electrolyte decomposition and hydrogen formation. Hydrogen evolution can cause internal pressure build-up, potentially leading to delamination, loss of interfacial contact, mechanical failure, and ultimately cell failure. Therefore, maintaining the operating voltage within the electrochemical stability window of the electrolyte, or using protective coatings at critical interfaces, is essential. It is worth noting that pressure build-up due to gas or volatile decomposition products is a known issue not only in solid-state batteries but also in lithium-ion batteries with liquid electrolytes.

3.3. Electrochemical Oxidation of $Li_2B_{12}H_{12}$. To further explore the electrochemical stability of $Li_2B_{12}H_{12}$, we applied the same oxidation method to $Li_2B_{12}H_{12}$ as starting material by preparing a $Li_2B_{12}H_{12}$-carbon composite as a working electrode and by assembling $Li/Li_2B_{12}H_{12}/Li_2B_{12}H_{12}$-carbon/Pt cells. Note that the thermal decomposition of $Li_2B_{12}H_{12}$ is only observed in its high-temperature phase (i.e., T > 355 °C); therefore, we do not expect thermal decomposition during the drying process.[59,62] Figure 3a shows the galvanostatic charge curve of the cell and the subsequent constant voltage step at 5.0 V vs Li^+/Li at 120 °C. Similar to the $LiBH_4$ oxidation discussed above, the voltage plateau starting at ~3.8 V vs Li^+/Li coincides with the onset of anodic currents in the cyclic voltammogram measured in the same cell setup at 120 °C (see Figure S7b), evidencing the electrochemical oxidation of $Li_2B_{12}H_{12}$.[19] The corresponding impedance spectra before the galvanostatic charge and after the constant voltage step are shown in Figure 3b. After oxidation, the bulk impedance at high frequencies increased by >80% together with the evolution of a semicircle (= interfacial impedance) of >250 kΩ at low frequencies, implying that the electrochemical oxidation of $Li_2B_{12}H_{12}$ decreases ionic conductivity and produces a highly resistive interphase.

The electrochemically oxidized composite in the working electrode was further characterized by B 1s XPS and ^{11}B NMR, as shown in Figure 3c,d, compared to the reference materials. Following electrochemical oxidation, the XPS spectrum of the $Li_2B_{12}H_{12}$ composite exhibits a distinct shoulder with a higher binding energy exceeding 189 eV compared to $Li_2B_{12}H_{12}$ alone. This suggests a partial transition to a higher boron oxidation state due to removing the valence electron charge. Moreover, the XPS spectrum and the positions and line shapes of the ^{11}B NMR resonances in the $Li_2B_{12}H_{12}$ composite after oxidation closely resemble those of a dehydrated $Li_3B_{24}H_{23}$ reference sample. However, distinguishing between peaks originating from residual $Li_2B_{12}H_{12}$ and the icosahedral B_{12} units in $Li_3B_{24}H_{23}$ presents a challenge. According to the

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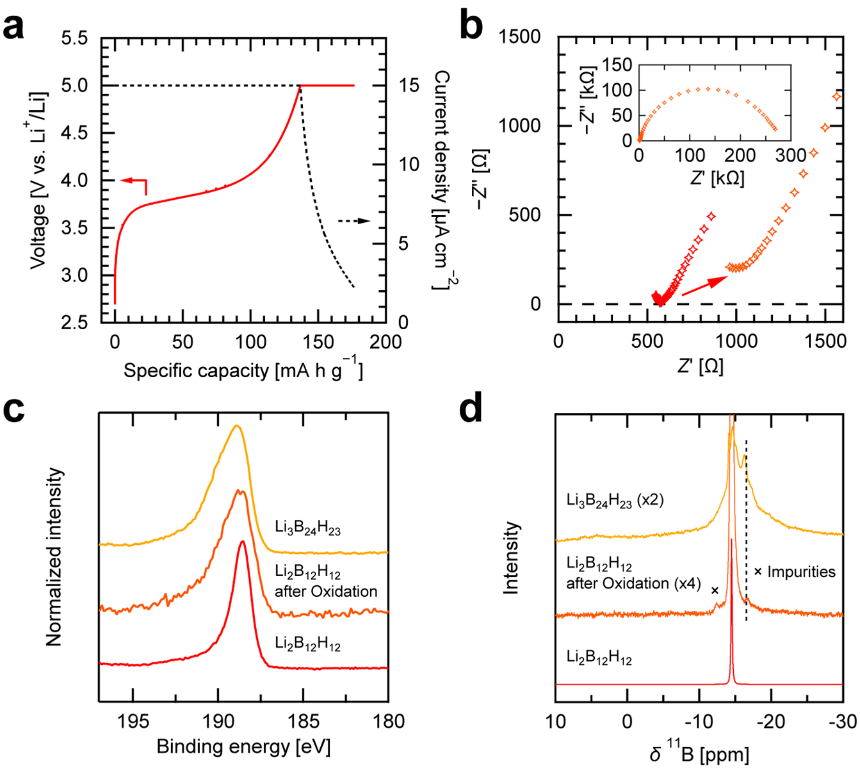

Figure 3. (a) Galvanostatic charge curve of electrochemical $Li_2B_{12}H_{12}$ oxidation at a current density of 15 μA cm^{-2} in a Li/$Li_2B_{12}H_{12}$/$Li_2B_{12}H_{12}$-carbon/Pt cell ($Li_2B_{12}H_{12}$:carbon = 75:25 in weight) at 120 °C. When the cell voltage reached 5.0 V vs Li^+/Li, the voltage was held constant until the current density dropped to <15% of the applied constant current density. The specific capacity was calculated based on the weight of $Li_2B_{12}H_{12}$ in the composite (= 3.75 mg). (b) Impedance spectra recorded before the galvanostatic charge (red) and after the constant voltage step (orange). The inset in (b) shows the impedance spectra over the full frequency range. (c) B 1s XPS and (d) ^{11}B NMR spectra of $Li_2B_{12}H_{12}$ (red), $Li_2B_{12}H_{12}$ after electrochemical oxidation (orange), and $Li_3B_{24}H_{23}$ (yellow).

former ab initio calculation, one possibility to explain the newly evolved peak at −16.5 ppm in the NMR spectrum can be attributed to a boron atom of B−H agglomerates;[57] however, their size and bonding remain unclear. These results indicate that $Li_2B_{12}H_{12}$ is electrochemically oxidized to larger agglomerates (dimers, trimers, and perhaps larger entities with several possible bonds) that are electrochemically more stable than $Li_2B_{12}H_{12}$, as supported by the stability window calculations (see Figure 1a).

Additionally, as shown in Figure 3b, the significant increase in bulk and interfacial impedance following the oxidation of $Li_2B_{12}H_{12}$ suggests that the electrochemically oxidized $Li_2B_{12}H_{12}$ no longer functions as a fast lithium-ion conductor. This behavior contrasts with that observed during the oxidation of $LiBH_4$. To achieve stable cell cycling with a $Li_2B_{12}H_{12}$ solid electrolyte, applying electrochemical potentials exceeding its oxidative stability limit of ∼3.8 V vs Li^+/Li should be avoided. Coupling with 4 V-class cathode active materials may still be possible by incorporating an interface protection layer (i) between the electrolyte and the active material and (ii) between the electrolyte and the electronically conductive additive in the cathode composite.[63] For example, titania protection layers have been successfully applied in the hydridoborate/NMC811 interface in the cathode composite.[30] Alternatively, combining two hydridoborates anions with distinct electrochemical stability limits proves to be an effective approach for extending the stability window, preserving the ionic conductivity of hydridoborate-based solid electrolytes. This occurs because the more stable anion component can continue to facilitate the ionic conduction, even as the less stable component begins to degrade beyond its stability threshold.[28] Current literature reports such extended electrochemical stability limits in Li or Na salts of mixed anions with *closo*-hydridoborate dimers,[38−40] *nido*-hydridoborates,[64,65] or *closo*-monocarba-hydridoborates,[28,30] suggesting that this is a generic phenomenon regardless of the types of hydridoborate anion clusters.

4. CONCLUSIONS

The electrochemical stability windows of $LiBH_4$ and $Li_2B_{12}H_{12}$ were calculated using a grand canonical method, including noncrystalline decomposition products. The combination of calculations with experiments on the solid-state electrochemical oxidation of hydridoborates reveals the following electrochemical oxidation pathway:

$LiBH_4 \rightarrow Li_2B_{12}H_{12}$

→ Li salts of longer H-interconnected *closo*-cluster anions

The proposed solid-state decomposition pathway clarifies the role of electrochemical reactions of $LiBH_4$ and $Li_2B_{12}H_{12}$ as solid electrolyte components on the cycle stability and compatibility with cathode active materials at high electrochemical potentials. Additionally, the emergence of a highly conductive $[BH_4]^-$−$[B_{12}H_{12}]^{2-}$ mixed phase, formed through the partial decomposition of $LiBH_4$, highlights the potential for *in situ* electrochemical preparation of ion-conductive mixed-anion hydridoborates within all-solid-state cells.

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While electrochemical decomposition in the solid state differs in conditions from thermal decomposition in vacuum or solution, previous studies have shown that both processes can lead to the formation of larger boron-containing species, ultimately resulting in amorphous boron. Similar trends have also been observed in solution-based studies, where cluster growth precedes complete decomposition.[16,66,67]

A comparison between theory and experiment reveals a difference in the predicted stability window for $Li_2B_{12}H_{12}$. The calculations for $LiBH_4$ align closely with experimental results, largely due to the well-characterized decomposition products, whereas the predictions for $Li_2B_{12}H_{12}$ deviate from what is observed experimentally. This could be due to uncertainties regarding the actual decomposition products formed during experiments, which might differ from those assumed in the theoretical model. Additionally, although the predicted decomposition products are thermodynamically feasible, their formation could be kinetically hindered, as suggested in ref 27 effectively extending the apparent experimental stability window in all-solid-state cells. The high voltage oxidation products are typically neither redox-active nor ionically conductive as the anion/cation ratio becomes unfavorable and anion rotational dynamics is sterically restricted, particularly in compounds with larger anions. For instance, ref 28 shows that cycling beyond the stability window leads to increasing impedance and eventual cell failure. Nevertheless, the overarching trend that larger clusters formed through oxidation are more stable at higher electrochemical potentials remains consistent across both theoretical predictions and experimental findings.

The calculations suggest that upon oxidation, $[B_{12}H_{12}]^{2-}$ anions form chains interconnected by bridging hydrogen atoms, which may contribute to the formation of a high-resistance layer at the cathode. Our computational approach and insights into electrochemical stability windows are broadly applicable to other hydridoborate systems, independent of the specific cation or anion structure. Based on observations from thermal decomposition studies with other cations such as Na^+ and Ca^{2+}, we anticipate that similar oxidation cascades, involving cluster growth and eventual formation of amorphous boron, may also occur in sodium-based and divalent systems such as $NaBH_4$, $Na_2B_{12}H_{12}$, and $Ca(BH_4)_2$. Thus, the methodology and conclusions presented here are not limited to lithium systems but may inform the broader design of alkali and alkaline earth metal-based solid-state batteries. In conclusion, this work provides general insights into the solid-state electrochemical decomposition behaviors of hydridoborate anions at the solid–solid interfaces to determine cathode design strategies, facilitating the development of high-voltage all-solid-state batteries using hydridoborate-based solid electrolytes.

■ ASSOCIATED CONTENT

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsaem.5c01264.

Supplementary notes on the electrochemical stability window calculations and electronic properties, DSC-TG-MS data of $Li_3B_{24}H_{23}\cdot xH_2O$, cyclic voltammograms of $LiBH_4$ and $Li_2B_{12}H_{12}$, synchrotron XRD patterns of $LiBH_4$, oxidized $LiBH_4$, and $Li_2B_{12}H_{12}$, structures of the molecular and cluster systems (PDF)

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Notes

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